

ACTIVITY 2 — FACILITATOR GUIDE

Measure — Map the Process and Spot the Waste

TYPE	DURATION	MAPS TO	TOPIC
Hands-on Lab	30 minutes	MEASURE	Measure — See What Is Really Happening

Purpose of this activity

By the end of this activity learners will be able to describe how the process runs and identify the eight wastes in it (A4, K2).

What learners produce

A completed SIPOC, a 15-step process map with each step judged VA or NVA, a completed DOWNTIME waste tally, and the written baseline figures.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- Flipchart or A3 sheet and markers
- The course slides for the scenario and framework
- Internet access where the activity calls for live research

Set-up

Groups of 3–5. Distribute one worksheet per learner and one checklist per group, and make sure every group can open this activity's data/ folder. All 5 activities run on the same continuous BrewBean Cafe scenario and follow DMAIC in order, so remind learners where this activity sits in that story — and that its output feeds the next one.

Scenario / briefing material

The team now has an agreed problem statement, but still only opinions about where the time goes. Management wants to act immediately; the team has persuaded them to spend one morning measuring first.

The improvement team spent one weekday morning observing the 7:30am-9:00am rush with a stopwatch and a check sheet. They timed 60 customers end to end, recorded what each one ordered, how long they waited, whether the drink had to be remade, and what was visibly holding things up. They also timed every step of the process itself.

Your job is to turn that raw observation into a picture of the process as it really runs — not as the staff handbook describes it — and to find where the time and effort are actually being lost.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Agree the start and stop points. Fix the boundaries before anything else: start when the customer joins the queue, stop when the customer receives the drink. A process with no agreed boundaries produces a map nobody can compare against anything.	Hold the line at 5-7 steps in the SIPOC Process column. Tables will want to list all fifteen; that is the process map's job.
2	Complete the SIPOC. Fill templates/sipoc-template.csv: Suppliers, Inputs, Process, Outputs, Customers. Hold the Process column to 5-7 high-level steps — the detail belongs in the process map, not here. A SIPOC is the one-page view a manager can read in thirty seconds.	Watch for learners marking the whole process value-adding because 'the customer wants coffee'. Re-apply the test to each individual step.
3	Build the process map with real timings. Open data/process-step-timings.csv — the 15 observed steps with their average times. Copy them into templates/process-map-template.csv. Note the 'Times Step Was Skipped' column: step 10 (fetching milk) happened for 18 of 60 customers and step 15 (remaking a drink) for 7 of 60.	'Drink Ordered' is categorical, not continuous. Steer them to the two types the course teaches and note that a label is neither.
4	Judge every step value-adding or not. Mark each step VA or NVA and write one line saying why. The test is applied per STEP: would the customer pay extra for THIS step? Not 'does the customer want coffee'. Expect to find that the great majority of the elapsed time is non-value-adding — that is the normal and useful finding.	Some tables tally waste from memory. Point them back to the data — the tally must be evidence, not impression.
5	Classify the data you have. Look at the columns in data/morning-rush-observation-log.csv. Which figures are discrete (counted — remakes, tallies) and which are continuous (measured on a scale — wait time in minutes)? Note that 'Drink Ordered' and 'Delay Reason' are neither: they are categorical labels.	Park any 'hire more staff' suggestions. Solutions are Lab 4; this phase is facts.
6	Complete the DOWNTIME waste tally. Open data/downtime-waste-tally.csv — three rows are completed as worked examples. Using the observation log and the step timings as your EVIDENCE, complete the remaining five rows in templates/downtime-tally-template.csv. Some waste types may legitimately be zero here; recording zero with evidence is a correct answer.	

7	Write down the baseline. Record the average wait, the longest wait, the percentage of customers over the 5-minute promise, and the number of drinks remade. Identify which single waste type you tallied most often and at which step. Lab 5 measures the improvement against these numbers, and the Case Study expects a baseline to be quoted rather than invented.	
---	--	--

Suggested timing

Minutes	Phase
5 min	Agree the process boundaries and complete the SIPOC
9 min	Build the process map and judge each step value-adding or not
4 min	Classify the data as discrete or continuous
8 min	Complete the DOWNTIME waste tally from the evidence
4 min	Write down the baseline figures

Total: 30 minutes

Discussion & decision prompts

1. Looking at your process map, what proportion of the total time is genuinely value-adding — and does that surprise you?
2. Which single step would you attack first, and is that step a symptom or a cause?
3. Which waste types came out at zero, and what evidence lets you say so with confidence?
4. Why does the team need to agree what is recorded, who records it and when, BEFORE collection starts?

Debrief

Bring the class back together and draw out the learning:

- Ask two or three groups to show what they produced, not just what they concluded.
- Ask what surprised them — the gap between what they assumed and what they found.
- Link the activity explicitly back to DMAIC · MEASURE — Measure — See What Is Really Happening.
- Remind learners this activity rehearses what the Case Study assessment will require.

Reflection questions:

- Was the step you assumed was the bottleneck actually the one the data pointed to?
- How would the map differ if you had asked staff to describe the process instead of observing it?
- Which of these wastes do you recognise in your own workplace?

Facilitator notes — what to watch for

- Hold the line at 5-7 steps in the SIPOC Process column. Tables will want to list all fifteen; that is the process map's job.
- Watch for learners marking the whole process value-adding because 'the customer wants coffee'. Re-apply the test to each individual step.

- 'Drink Ordered' is categorical, not continuous. Steer them to the two types the course teaches and note that a label is neither.
- Some tables tally waste from memory. Point them back to the data — the tally must be evidence, not impression.
- Park any 'hire more staff' suggestions. Solutions are Lab 4; this phase is facts.

Success criterion

Every group can show a SIPOC with all five columns filled, a process map with every step timed and judged VA or NVA, a complete DOWNTIME tally tied to evidence, and the written baseline figures.

Assessment link

This activity builds MEASURE — Measure — See What Is Really Happening. All 5 activities are in-class activities assessed indirectly through the open-book Case Study (CS) component, which reuses the same techniques on the same continuous BrewBean Cafe scenario. Knowledge is assessed in the Written Assessment (WA-SAQ). Nothing is assessed that is not practised here or taught in the slides.